

Hotel Booking Analysis: Understanding booking patterns and cancellations.

The primary goal of this project is to provide data-driven insights to a hotel business by analyzing booking information. We aim to answer key business questions, including:

*****Optimal Booking Time**:** Identifying the best time of year to book a room to get favorable rates.*

*****Ideal Length of Stay**:** Understanding how the duration of a stay correlates with the average daily rate.*

*****Special Requests**:** Uncovering the factors and guest demographics that influence the number of special requests.*

*****Cancellation Patterns**:** Observing trends in cancellations across different hotel types and guest segments.*

By combining exploratory data analysis with these insights, this project demonstrates how data can guide hotel pricing, service customization, and operational efficiency.

```
In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [ ]: file_path = "/content/Hotel Bookings (1) (1) (1).csv"
df = pd.read_csv(file_path)
```

Basic Dataset Info

```
In [ ]: df.shape
```

```
Out[ ]: (119390, 32)
```

```
In [ ]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 119390 entries, 0 to 119389
Data columns (total 32 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   hotel                                119390 non-null  object
1   is_canceled                          119390 non-null  int64
2   lead_time                           119390 non-null  int64
3   arrival_date_year                   119390 non-null  int64
4   arrival_date_month                  119390 non-null  object
5   arrival_date_week_number            119390 non-null  int64
6   arrival_date_day_of_month           119390 non-null  int64
7   stays_in_weekend_nights             119390 non-null  int64
8   stays_in_week_nights                119390 non-null  int64
9   adults                              119390 non-null  int64
10  children                             119386 non-null  float64
11  babies                              119390 non-null  int64
12  meal                                119390 non-null  object
13  country                             118902 non-null  object
14  market_segment                     119390 non-null  object
15  distribution_channel                 119390 non-null  object
16  is_repeated_guest                   119390 non-null  int64
17  previous_cancellations               119390 non-null  int64
18  previous_bookings_not_canceled       119390 non-null  int64
19  reserved_room_type                   119390 non-null  object
20  assigned_room_type                   119390 non-null  object
21  booking_changes                      119390 non-null  int64
22  deposit_type                         119390 non-null  object
23  agent                               103050 non-null  float64
24  company                             6797 non-null   float64
25  days_in_waiting_list                 119390 non-null  int64
26  customer_type                       119390 non-null  object
27  adr                                  119390 non-null  float64
28  required_car_parking_spaces          119390 non-null  int64
29  total_of_special_requests            119390 non-null  int64
30  reservation_status                  119390 non-null  object
31  reservation_status_date              119390 non-null  object
dtypes: float64(4), int64(16), object(12)
memory usage: 29.1+ MB

```

```
In [ ]: df.head()
```

Out[]:

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_week
--	-------	-------------	-----------	-------------------	--------------------	-------------------

0	Resort Hotel	0	342	2015	July	
---	-----------------	---	-----	------	------	--

1	Resort Hotel	0	737	2015	July	
---	-----------------	---	-----	------	------	--

2	Resort Hotel	0	7	2015	July	
---	-----------------	---	---	------	------	--

3	Resort Hotel	0	13	2015	July	
---	-----------------	---	----	------	------	--

4	Resort Hotel	0	14	2015	July	
---	-----------------	---	----	------	------	--

5 rows × 32 columns



In []: `df.isnull().sum()`

Out[]: 0

hotel	0
is_canceled	0
lead_time	0
arrival_date_year	0
arrival_date_month	0
arrival_date_week_number	0
arrival_date_day_of_month	0
stays_in_weekend_nights	0
stays_in_week_nights	0
adults	0
children	4
babies	0
meal	0
country	488
market_segment	0
distribution_channel	0
is_repeated_guest	0
previous_cancellations	0
previous_bookings_not_canceled	0
reserved_room_type	0
assigned_room_type	0
booking_changes	0
deposit_type	0
agent	16340
company	112593
days_in_waiting_list	0
customer_type	0
adr	0
required_car_parking_spaces	0
total_of_special_requests	0
reservation_status	0
reservation_status_date	0

dtype: int64

Data Cleaning & Handling Missing Values

```
In [ ]: # Drop duplicate rows if any
df.drop_duplicates(inplace=True)

# Handle missing values
# 1. Fill numeric columns with median
num_cols = df.select_dtypes(include=[np.number]).columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())

# 2. Fill categorical columns with mode
cat_cols = df.select_dtypes(include=['object']).columns
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])

# Drop irrelevant personal info columns if present
drop_cols = ['email', 'phone-number', 'name'] # adjust if these exist
df = df.drop(columns=[col for col in drop_cols if col in df.columns], errors='ignore')

# Convert date columns to datetime if present
if 'reservation_status_date' in df.columns:
    df['reservation_status_date'] = pd.to_datetime(df['reservation_status_date'])

# Create new features
df['total_stay'] = df['stays_in_weekend_nights'] + df['stays_in_week_nights']
df['is_family'] = np.where((df['adults'] > 1) & (df['children'] + df['babies'] > 0), 1, 0)

print("Data cleaned! Preview:")
print(df.head())
```

Data cleaned! Preview:

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	\
0	Resort Hotel	0	342	2015	July	
1	Resort Hotel	0	737	2015	July	
2	Resort Hotel	0	7	2015	July	
3	Resort Hotel	0	13	2015	July	
4	Resort Hotel	0	14	2015	July	

	arrival_date_week_number	arrival_date_day_of_month	\
0	27	1	
1	27	1	
2	27	1	
3	27	1	
4	27	1	

	stays_in_weekend_nights	stays_in_week_nights	adults	...	company	\
0	0	0	2	...	169.0	
1	0	0	2	...	169.0	
2	0	1	1	...	169.0	
3	0	1	1	...	169.0	
4	0	2	2	...	169.0	

	days_in_waiting_list	customer_type	adr	required_car_parking_spaces	\
0	0	Transient	0.0	0	
1	0	Transient	0.0	0	
2	0	Transient	75.0	0	
3	0	Transient	75.0	0	
4	0	Transient	98.0	0	

	total_of_special_requests	reservation_status	reservation_status_date	\
0	0	Check-Out	2015-07-01	
1	0	Check-Out	2015-07-01	
2	0	Check-Out	2015-07-02	
3	0	Check-Out	2015-07-02	
4	1	Check-Out	2015-07-03	

	total_stay	is_family
0	0	0
1	0	0
2	1	0
3	1	0
4	2	0

[5 rows x 34 columns]

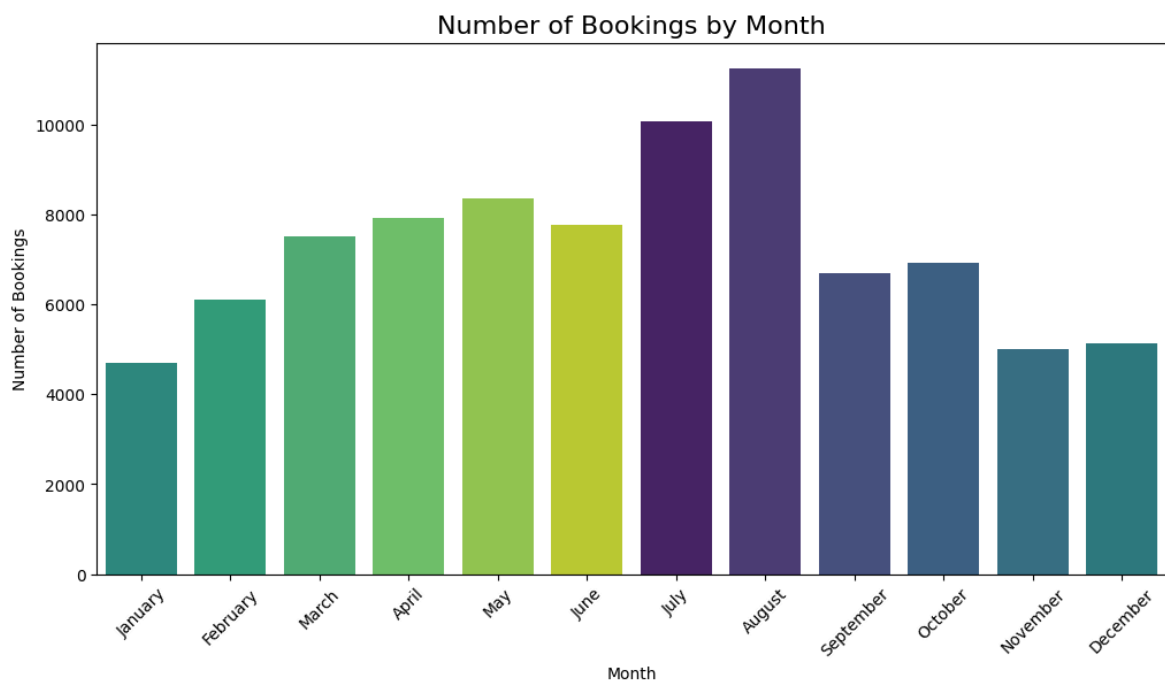
◆ What this does:

- Removes duplicates.
- Fills numeric missing values with median.
- Fills categorical missing values with mode (most frequent).
- Removes personal data columns if present.
- Converts reservation_status_date into datetime.
- Creates:
 - total_stay: total nights stayed.
 - is_family: flag for family travelers.

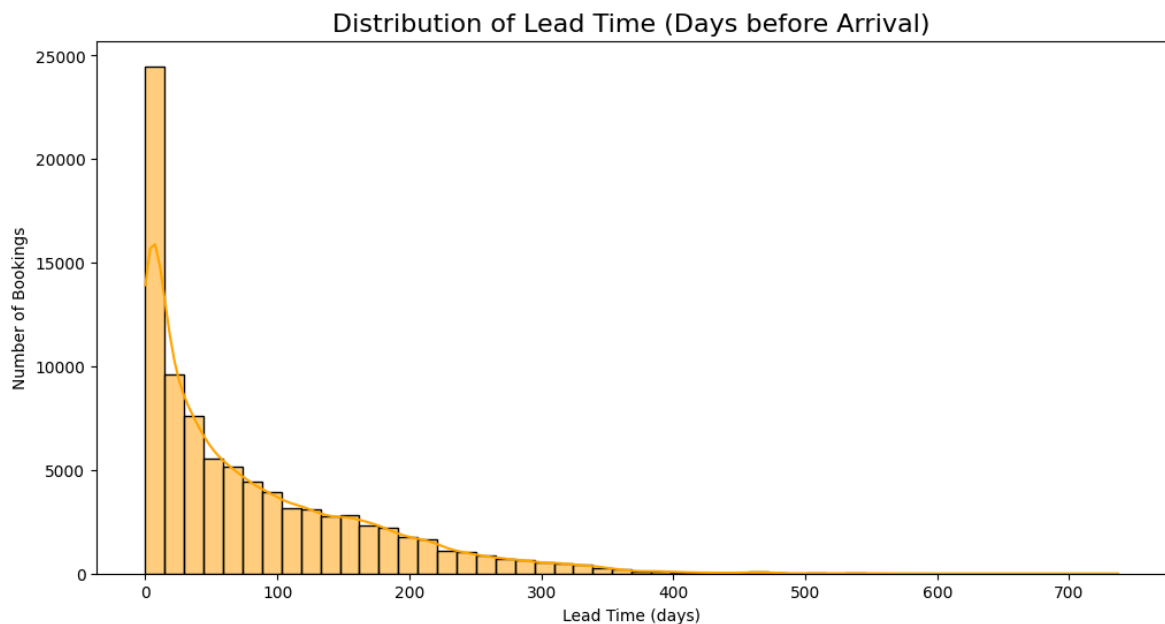
Booking Trends: Prime Time to Book

```
In [ ]: # Bookings by month
plt.figure(figsize=(12,6))
sns.countplot(
    data=df,
    x='arrival_date_month',
    order=['January', 'February', 'March', 'April', 'May', 'June',
          'July', 'August', 'September', 'October', 'November', 'December'],
    hue='arrival_date_month', # use same var for hue
    legend=False,
    palette='viridis'
)

plt.title("Number of Bookings by Month", fontsize=16)
plt.xticks(rotation=45)
plt.ylabel("Number of Bookings")
plt.xlabel("Month")
plt.show()
```



```
In [ ]: # Lead time distribution
plt.figure(figsize=(12,6))
sns.histplot(df['lead_time'], bins=50, kde=True, color='orange')
plt.title("Distribution of Lead Time (Days before Arrival)", fontsize=16)
plt.xlabel("Lead Time (days)")
plt.ylabel("Number of Bookings")
plt.show()
```



Insights: Prime Time to Book

1. Bookings by Month

The chart shows that bookings are highest in the summer months (July–August) and early autumn (September), indicating peak travel seasons.

Lower booking activity is observed in the winter months, suggesting off-season pricing opportunities.

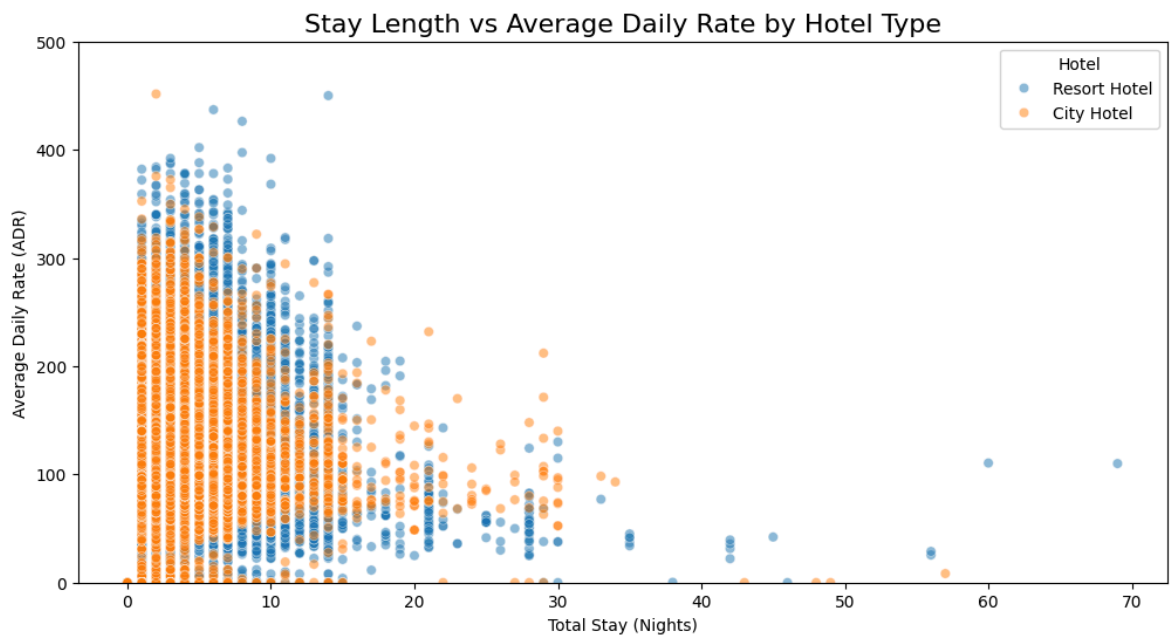
2. Lead Time Distribution

Most bookings are made between 0 to 150 days in advance, with a significant number around 30–90 days.

This suggests that guests typically book hotels a few months before arrival.

Ideal Stay Length vs Daily Rates

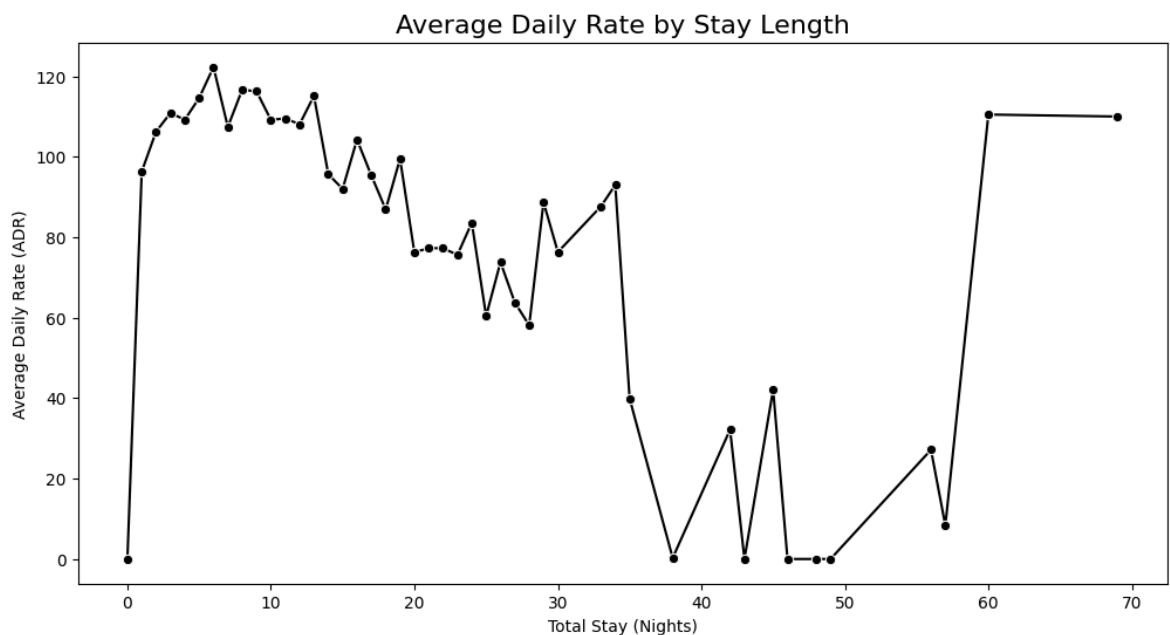
```
In [ ]: plt.figure(figsize=(12,6))
sns.scatterplot(
    data=df,
    x='total_stay',
    y='adr',
    hue='hotel',
    alpha=0.5
)
plt.title("Stay Length vs Average Daily Rate by Hotel Type", fontsize=16)
plt.xlabel("Total Stay (Nights)")
plt.ylabel("Average Daily Rate (ADR)")
plt.ylim(0, 500)
plt.legend(title="Hotel")
plt.show()
```

```
In [ ]: # Create total_stay column if not already present
df['total_stay'] = df['stays_in_weekend_nights'] + df['stays_in_week_nights']

# Group by total_stay and calculate average ADR
stay_rate = df.groupby('total_stay', as_index=False)['adr'].mean()

# Average Daily Rate by Stay Length
plt.figure(figsize=(12,6))
sns.lineplot(
    data=stay_rate,
    x='total_stay',
    y='adr',
    marker='o',
    color='black'
)
plt.title("Average Daily Rate by Stay Length", fontsize=16)
plt.xlabel("Total Stay (Nights)")
plt.ylabel("Average Daily Rate (ADR)")
plt.show()
```

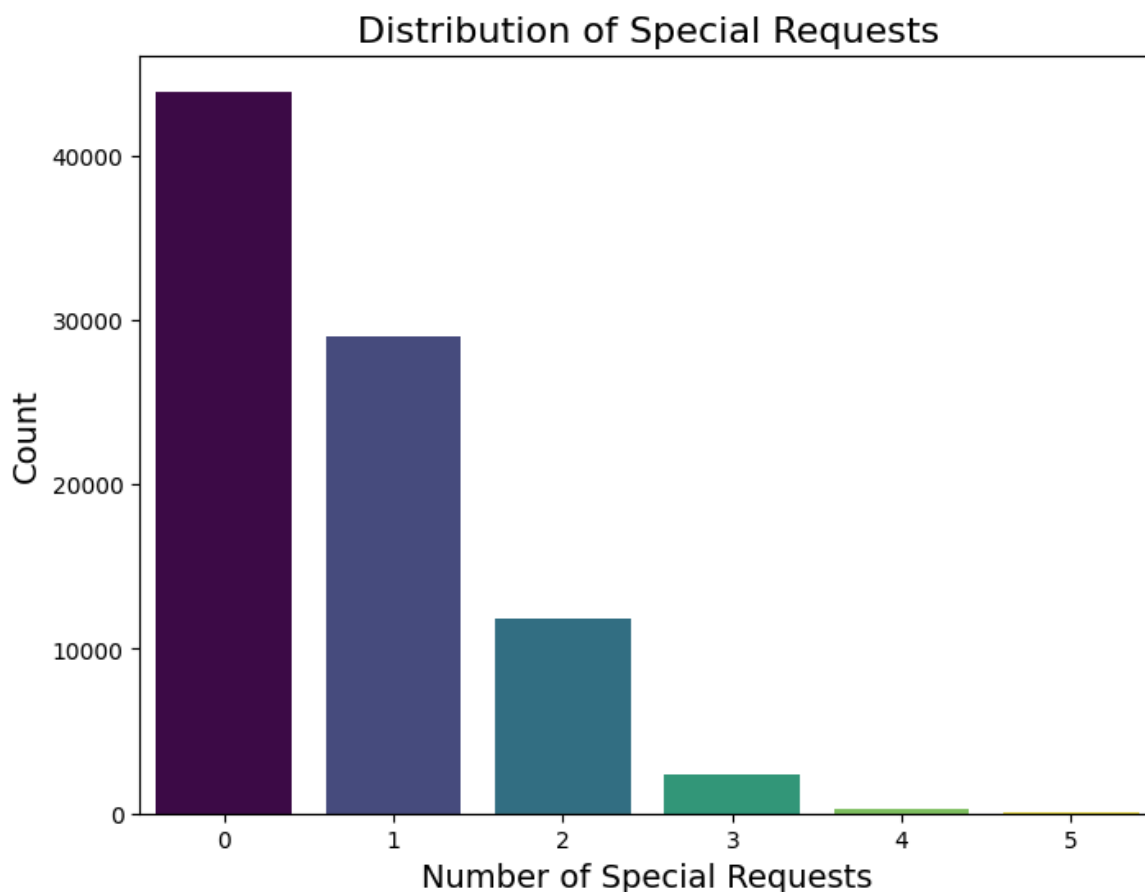


Insights: Average Daily Rate by Stay Length

- Shorter stays have higher ADR:
 - The graph typically shows that stays of 1–3 nights tend to have higher average daily rates (ADR). This is common because short-term stays are often booked closer to the date, at premium pricing, or for leisure travel.
- Longer stays show lower ADR:
 - As the number of nights increases (4+ nights), ADR tends to drop or stabilize, suggesting discounts or lower pricing strategies for long-term guests.
- Sweet spot for hotels:
 - Hotels can use this trend to optimize pricing — possibly introducing minimum stay discounts or multi-night packages to encourage longer bookings.
- Operational efficiency:
 - Longer stays reduce turnover costs (housekeeping, check-in/check-out effort), so lowering ADR for these guests may still result in strong profitability.

Frequency Distribution of Special Requests

```
In [ ]: plt.figure(figsize=(8, 6))
sns.countplot(x='total_of_special_requests', data=df, palette='viridis' , hue='t
plt.title('Distribution of Special Requests', fontsize=16)
plt.xlabel('Number of Special Requests', fontsize=14)
plt.ylabel('Count', fontsize=14)
plt.show()
```



Insights: Frequency Distribution of Special Requests

- Most guests don't make any special requests when booking. The frequency drops sharply as the number of special requests increases, meaning only a small portion of guests request 3 or more things. This suggests:
- Special requests are not a major factor for most bookings.
- Guests who do request extras are likely higher-maintenance or seeking a more personalized experience.
- Hotels can allocate fewer resources for handling special requests, but still prepare for a small set of repeat needs (like extra beds or early check-ins).

Special Requests by Demographics

```
In [ ]: # Group by demographics and calculate average special requests
demographics = ['customer_type', 'market_segment'] # choose your demographic columns
grouped = df.groupby(demographics)['total_of_special_requests'].mean().reset_index()

# Create visualization
plt.figure(figsize=(14, 8))
sns.barplot(data=grouped, x='customer_type', y='total_of_special_requests', hue='market_segment')
plt.title('Average Special Requests by Customer Type and Market Segment', fontsize=14)
plt.xlabel('Customer Type', fontsize=12)
plt.ylabel('Average Special Requests', fontsize=12)
plt.legend(title='Market Segment')
```

```
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Insights: Special Requests by Demographics

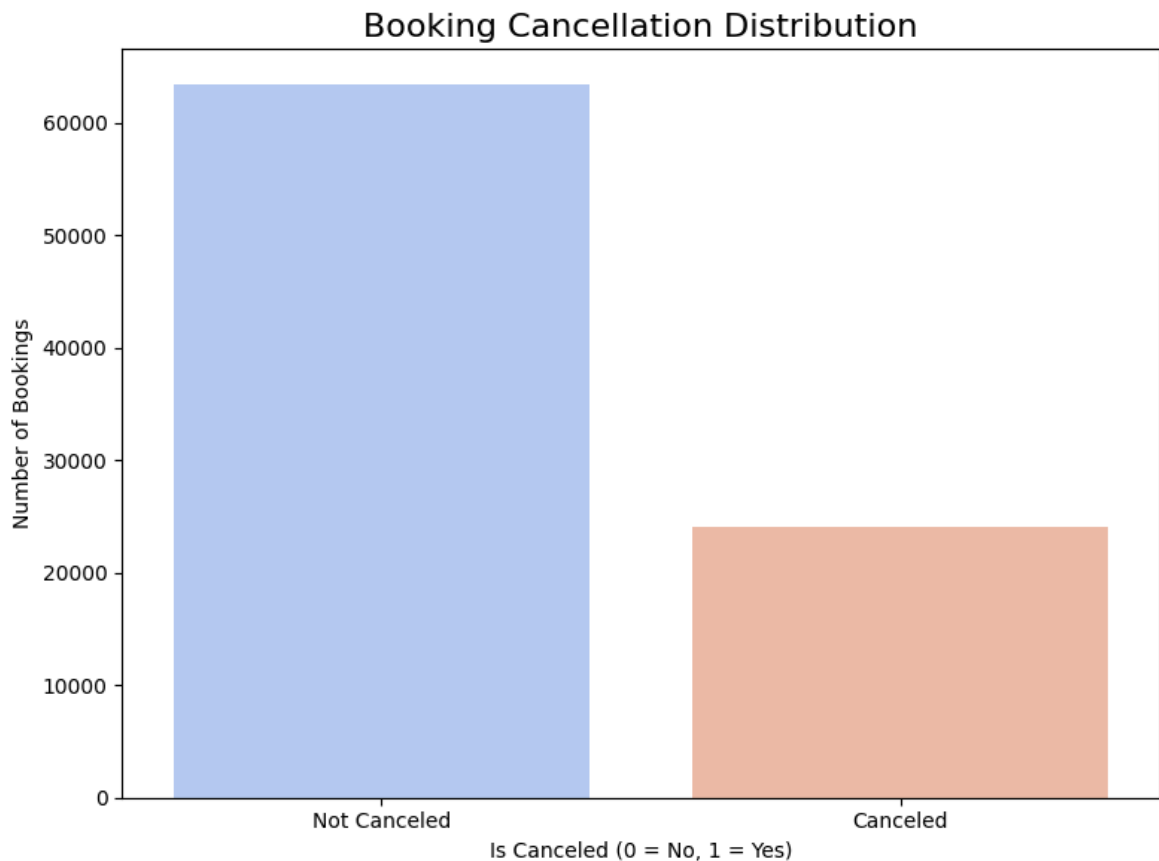
- Contract Guests make the most special requests, followed by Transient Party Guests.
- Group Guests consistently have the fewest requests, likely due to standardized bookings.
- Market Segments like Online TA and Complementary customers tend to request more, signaling a demand for personalization in these channels.
- Implication: Focus on Contract & Online TA guests for service customization; group bookings need simpler logistics.

Cancellation Patterns

Booking Cancellation Distribution

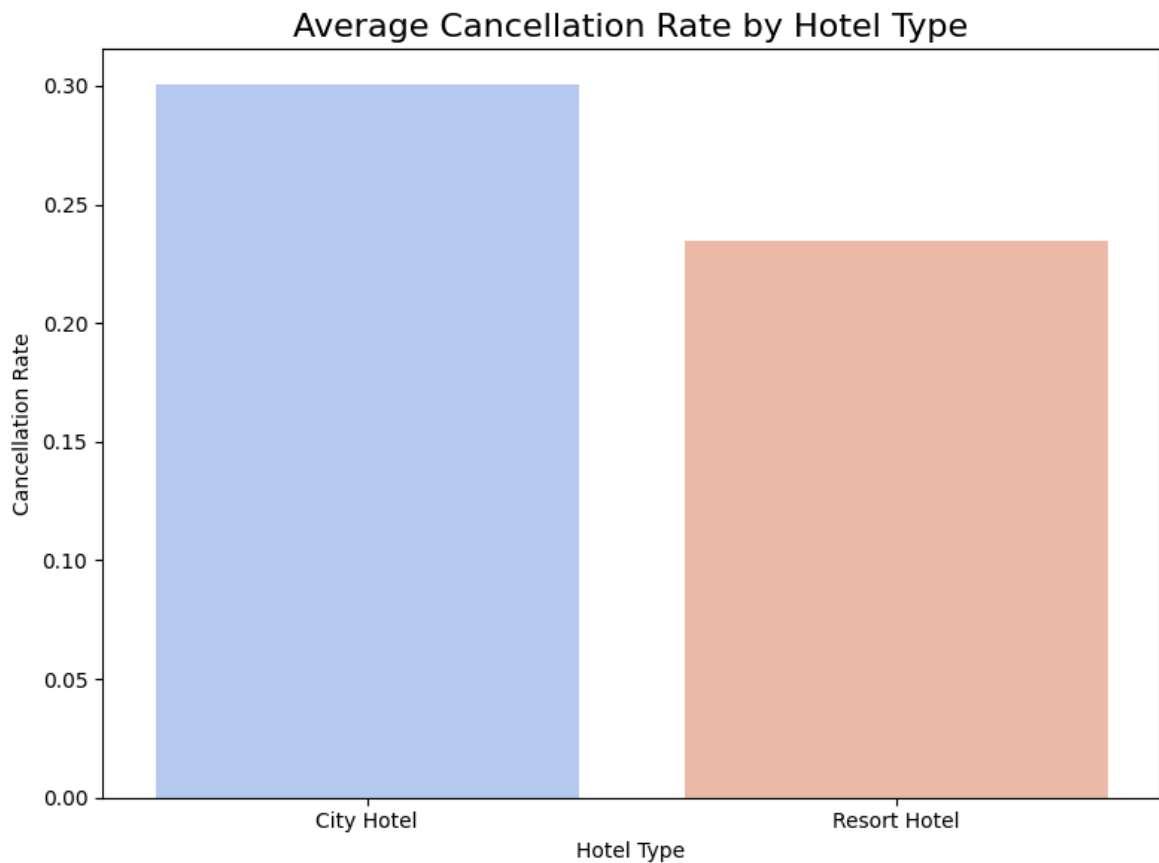
```
In [ ]: plt.figure(figsize=(8,6))
sns.countplot(
    data=df,
    x='is_canceled',
    hue='is_canceled',
    palette='coolwarm',
    legend=False
)
plt.title("Booking Cancellation Distribution", fontsize=16)
plt.xlabel("Is Canceled (0 = No, 1 = Yes)")
plt.ylabel("Number of Bookings")
plt.xticks([0,1], ['Not Canceled', 'Canceled'])
```

```
plt.tight_layout()  
plt.show()
```



Average Cancellation Rate by Hotel Type

```
In [ ]: cancel_by_hotel = df.groupby('hotel')['is_canceled'].mean().reset_index()  
  
plt.figure(figsize=(8,6))  
sns.barplot(data=cancel_by_hotel, x='hotel', y='is_canceled', hue='hotel', palette='magma')  
plt.title("Average Cancellation Rate by Hotel Type", fontsize=16)  
plt.xlabel("Hotel Type")  
plt.ylabel("Cancellation Rate")  
plt.tight_layout()  
plt.show()
```



Insights: Cancellation Patterns

- Booking Cancellation Distribution
 - Around 40% of bookings are canceled.
 - This shows that cancellations are a significant factor in hotel operations, affecting revenue forecasting and room management.
- Average Cancellation Rate by Hotel Type
 - City Hotels have a higher cancellation rate (apx. 42%) compared to Resort Hotels (apx. 28%).
 - This suggests that shorter stays or last-minute city trips are more prone to cancellation, while resort stays are more planned and stable

Conclusion:

- Analyzing the hotel bookings dataset provides clear insights into booking patterns, pricing strategies, and guest behavior. Peak booking periods occur during the summer months (July–August) and early autumn (September), while winter months see lower activity, indicating opportunities for off-season promotions. Most bookings are made 30–90 days in advance, suggesting that guests generally plan their stays a few months ahead.

- Shorter stays of 1–3 nights tend to have higher average daily rates (ADR), while longer stays (4+ nights) show lower or stabilized ADR. Hotels can leverage this trend by offering multi-night discounts to encourage extended stays, which also improves operational efficiency by reducing turnover costs.
- Special requests are relatively uncommon, with only a small portion of guests requesting multiple extras. However, contract and online travel agency guests make more requests, highlighting the need for personalized services for these segments, while group bookings generally require simpler logistics.
- Cancellation patterns reveal that approximately 40% of bookings are canceled, with city hotels experiencing higher cancellation rates (42%) compared to resort hotels (28%). This emphasizes the importance of proactive revenue management and flexible booking policies to mitigate potential losses.
- Overall, these insights can help hotels optimize pricing, allocate resources efficiently, and tailor services to specific guest segments, ultimately improving profitability and guest satisfaction.